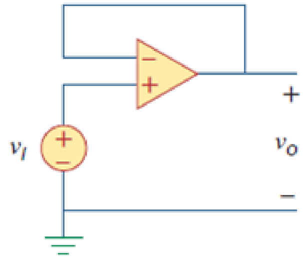
Chapter 5, Problem 5P

Problem

For the op amp circuit of Fig. 5.44, the op amp has an open-loop gain of 100,000, an input resistance of 10 kΩ, and an output resistance of 100 Ω. Find the voltage gain *vo/vi* using the nonideal model of the op amp.

Figure. 5.44:



Step-by-step solution

Step 1 of 5

Given that,

$R_i = 10\text{k}\Omega$,

$R_o = 100\Omega$ and

$A = 100,000$.

Step 2 of 5

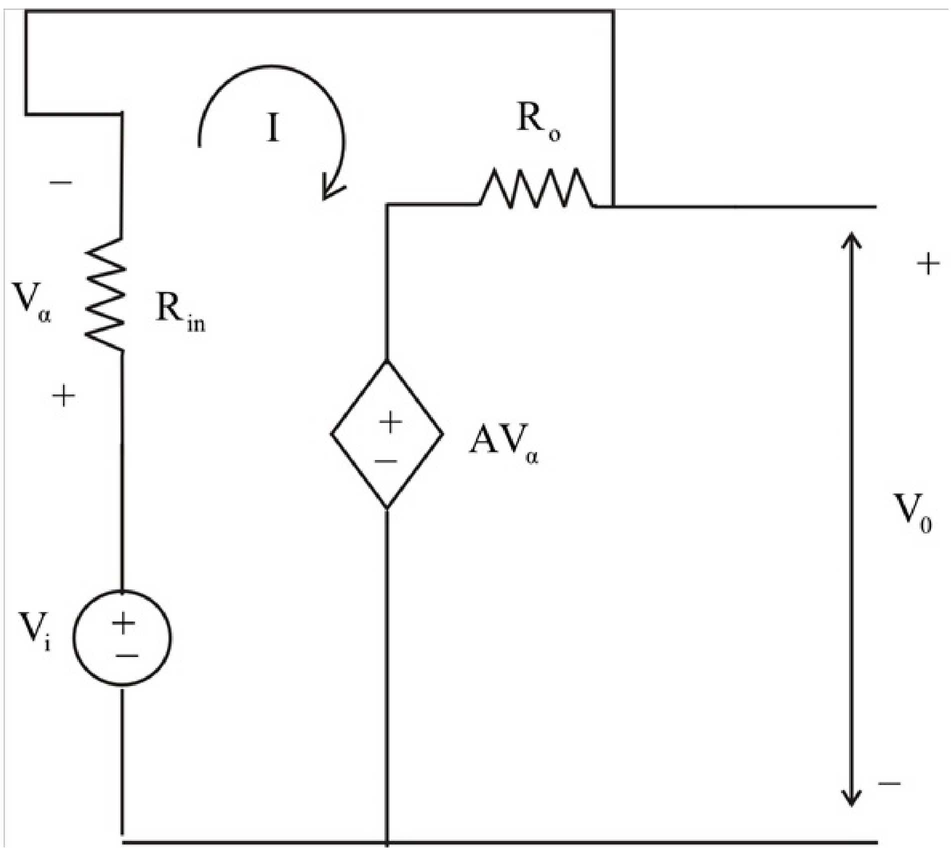


Fig-(a)

Step 3 of 5

From Fig-(a),

Applying KVL,

$-V_i + AV_\alpha + (R_i + R_0)I = 0$ .... (1)

But,

$V_\alpha = R_i I$ .... (2)

From Equations-(1)

$-V_i + (R_i + R_0 + R_i A)I = 0$

$I = \dfrac{V_i}{R_0 + (1+A)R_i}$ .... (3)

Step 4 of 5

Applying KVL to outer loop we get,

$-AV_\alpha - R_0 I + V_0 = 0$

$V_0 = AV_\alpha + R_0 I$

$V_0 = (R_0 + R_i A)I$ (From Equations-(2))

$V_0 = \dfrac{(R_0 + R_i A)V_i}{R_0 + (1+A)R_i}$ (From Equations-(3))

Step 5 of 5

Therefore,

$$\frac{V_0}{V_i} = \frac{R_0 + R_i A)}{R_0 + (1+A)R_i}$$

$$\frac{V_0}{V_i} = \frac{100 + 10\times10^3\times10^5}{100 + (1+10^5)\times10^4}$$

$$\frac{V_0}{V_i} \approx \frac{10^9}{(1+10^5)\times10^4}$$

$$\boxed{\frac{V_0}{V_i} = 0.9999990}$$

Comments (1)

Anonymous

the correct answer is 0.999990 not 0.9999990.